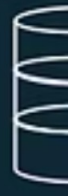


AWS Cloud Security Model & Shared Responsibility

Understanding how security responsibilities are divided between AWS and customers across different service models

Identity, and compliance		
 Detection	 Infrastructure protection	 Data protection
Security Hub GuardDuty Inspector CloudWatch Config CloudTrail Flow Logs	AWS Firewall Manager AWS Shield AWS WAF – Web application firewall Amazon Virtual Private Cloud (VPC) AWS PrivateLink AWS Systems Manager	Amazon S3 AWS Key Management Service AWS CloudHSM AWS Certificate Manager AWS Secrets Manager AWS IAM Server-Side Encryption

AWS Cloud Security Model

Security OF the Cloud

Managed by AWS - infrastructure, hardware, and facilities

Security IN the Cloud

Managed by customer - applications, data, and configurations

The security boundary shifts depending on the service model you choose. AWS manages the underlying infrastructure security, while customers control their applications and data. This fundamental division of responsibility varies across IaaS, PaaS, and SaaS services.

AWS operates a global infrastructure with comprehensive compliance certifications including SOC, ISO 27001, FedRAMP, PCI-DSS, and HIPAA. These certifications help customers meet their own regulatory requirements.

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Security of the Cloud

AWS Responsibility



Physical Security

Data centers with multiple layers of access control, environmental safeguards, and 24/7 monitoring



Virtualization Layer

Controlling the hypervisor, host operating system, and virtualization software



Hardware & Infrastructure

Managing physical servers, storage devices, and networking equipment across global regions



Maintenance & Patching

Automated patching and updates of underlying infrastructure components

AWS has invested billions in data center security, redundancy, and compliance infrastructure. Customers do not manage these layers - AWS handles all aspects of physical and infrastructure security.

Security in the Cloud

Customer Responsibility

Operating Systems

Patching and maintaining guest OS on services like EC2

Applications

Managing application code, dependencies, and updates

Data Protection

Implementing encryption, classification, and backup strategies

Identity & Access

Configuring IAM policies, roles, and user permissions

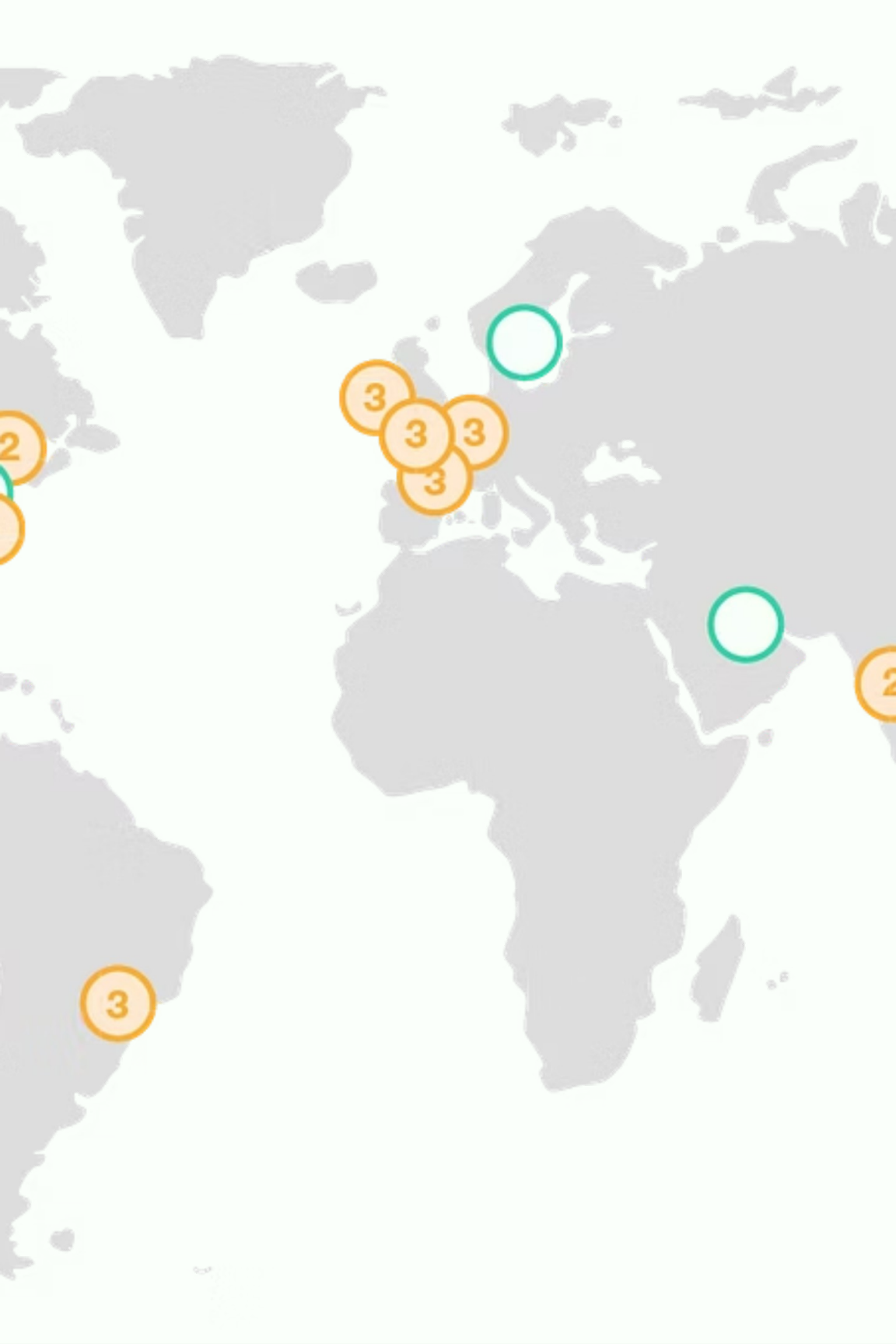
Network Security

Setting up security groups, NACLs, and firewall rules

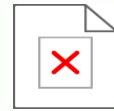
Monitoring & Logging

Tracking access, analyzing logs, and responding to incidents

Customer responsibilities vary significantly by service chosen. Misconfiguring customer-side controls is a common source of security breaches. Understanding your specific responsibilities is critical for maintaining a strong security posture.



AWS Global Infrastructure



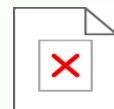
Worldwide Presence

AWS operates data centers across multiple continents with strict compliance to physical security standards



Multiple Control Layers

Physical security, environmental safeguards, and strict access control at every facility



Redundancy & Reliability

Built-in redundancy helps mitigate risks of data breaches and downtime

AWS Compliance & Certifications

AWS maintains compliance with major regulatory frameworks, significantly reducing customer compliance burden. These certifications demonstrate AWS's commitment to security and help customers meet their own regulatory requirements.



SOC 1/2/3

Service Organization Controls for financial and security audits



FedRAMP

Critical compliance for US federal agencies and government work



PCI-DSS

Payment card industry data security standards



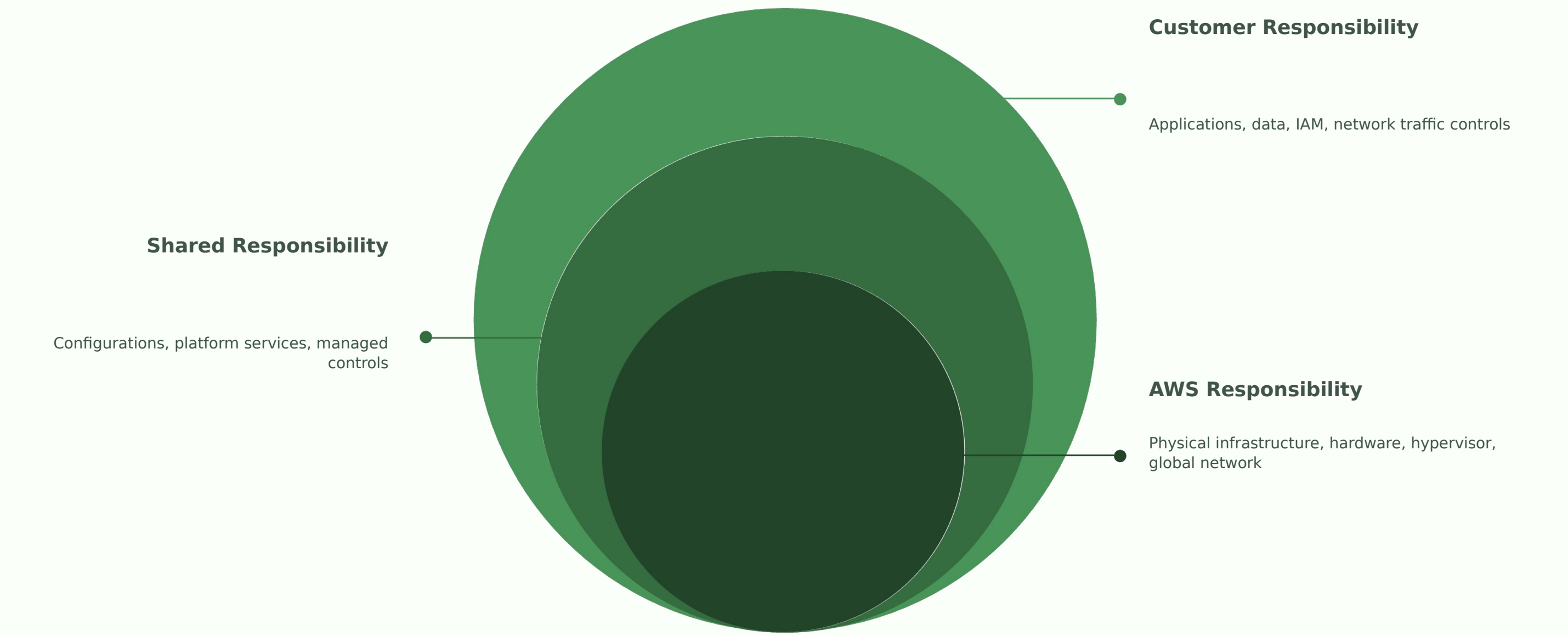
HIPAA

Healthcare information privacy and security requirements

GDPR

European data protection and privacy regulations

Deep Dive: Shared Responsibility Model



The shared responsibility model is the foundation of AWS security. Understanding this model is essential for all AWS users. It helps reduce operational burden while protecting resources effectively.

Customer vs AWS Control Areas

Customer Controls

- **Data & Applications**

Manage data, guest OS, and application software

- **Security Configuration**

Implement IAM policies and set firewall/security groups

- **Patching & Updates**

Patch applications and operating systems

- **Encryption Management**

Manage encryption keys and certificates

- **Monitoring**

Monitor and log access to resources

AWS Controls

- **Physical Security**

Controls physical security and storage hardware

- **Network Infrastructure**

Maintains global network infrastructure

- **Virtualization**

Manages hypervisor and virtualization layer

- **Infrastructure Patching**

Patches underlying infrastructure components

- **Facilities**

Manages power, cooling, and facility operations

The level of customer responsibility depends on the selected AWS service. For example, EC2 requires more customer management than Lambda or Fargate.

Service Comparison: EC2 vs RDS vs Lambda

This comparison demonstrates how responsibility shifts based on service abstraction level. The more managed the service, the more AWS handles infrastructure concerns.

Responsibility Area	EC2 (IaaS)	RDS (PaaS)	Lambda (FaaS)
OS Patch Management	✓ Customer	✓ AWS (mostly)	✓ AWS (fully)
Platform Maintenance	✓ Customer	✓ AWS	✓ AWS
Application Management	✓ Customer	✓ Customer	✓ Customer
Network Security	✓ Customer	✓ Customer	✓ Customer
Physical Security	AWS	AWS	AWS
Backup & Recovery	Customer	Shared	Shared

Understanding Service Abstraction



EC2 - IaaS

Infrastructure as a Service with highest customer responsibility for OS, patching, and maintenance



RDS - PaaS

Platform as a Service with shared responsibility - AWS manages engine patching, customer secures data



Lambda - FaaS

Function as a Service with lowest customer infrastructure responsibility - focus only on code

As you move from left to right, AWS takes on more infrastructure management responsibilities, allowing you to focus more on your application logic and business value.

Real-World Examples

1

EC2 Instance Patching

Reality: Customer owns the OS patches for EC2 instances. You must implement a patching strategy, test updates, and apply them to your instances.

AWS provides tools like Systems Manager Patch Manager, but the responsibility to patch remains with the customer.

2

RDS Database Management

Reality: AWS manages database engine patching automatically during maintenance windows. However, customers handle data security, access controls, encryption, and backup retention policies.

You configure security groups, IAM policies, and choose encryption options.

3

Lambda Function Security

Reality: AWS manages the OS and infrastructure completely in Lambda. There is no operating system to patch from the customer perspective - Lambda abstracts this away entirely.

Customers focus solely on function code, IAM roles, and environment variables.

Interactive Question

Who manages OS patching in Lambda?

Answer

AWS manages the OS and infrastructure completely in Lambda. There is no operating system to patch from the customer perspective. Lambda abstracts this away entirely, allowing you to focus purely on your application code.

Key Takeaway

Service abstraction shifts responsibility. The more managed the service, the less infrastructure management you need to handle.

Your Focus

With serverless services like Lambda, you can concentrate on building features and delivering business value instead of managing infrastructure.

